# Poverty amid Progress:
## The Hidden Cost of a Cold Shock in Mongolia

**Ade Febriady**
University of Groningen · World Bank
a.febriady@rug.nl (corresponding author)
**Lydia Kim**
World Bank

febriady.github.io

The 2023/24 dzud killed **nine million livestock, one in seven.** Yet **poverty kept falling fast.**

A dzud is an extreme winter of deep cold, heavy snow and ice-locked pasture. Outside the capital, one household in three herds.

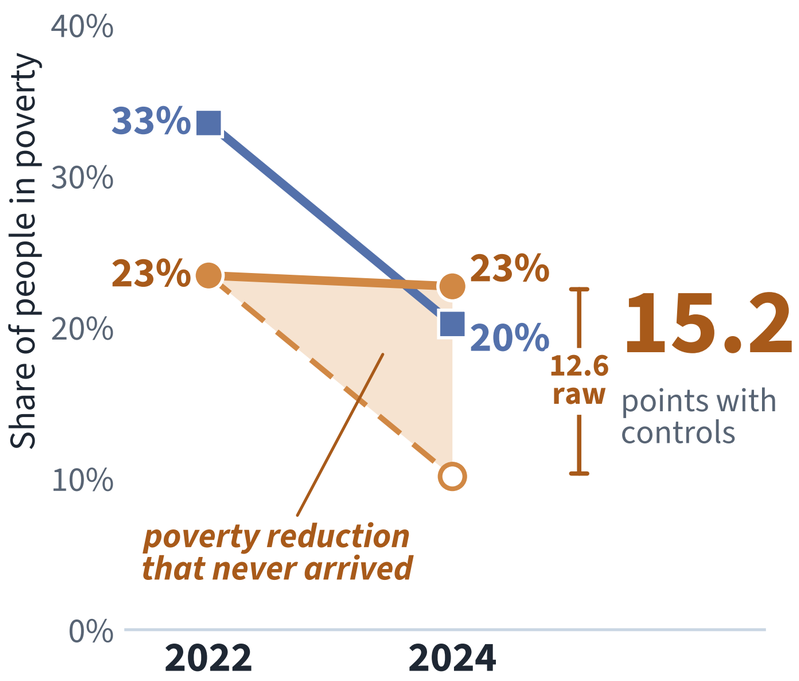
**What did falling poverty hide? For whom, and how?**
**A large, concentrated loss. Herders most, but not alone.**

Severely affected districts · Less affected districts · Severely affected, had poverty fallen like the rest

### Underneath the decline

#### All households

Flat where the winter hit hardest, falling elsewhere



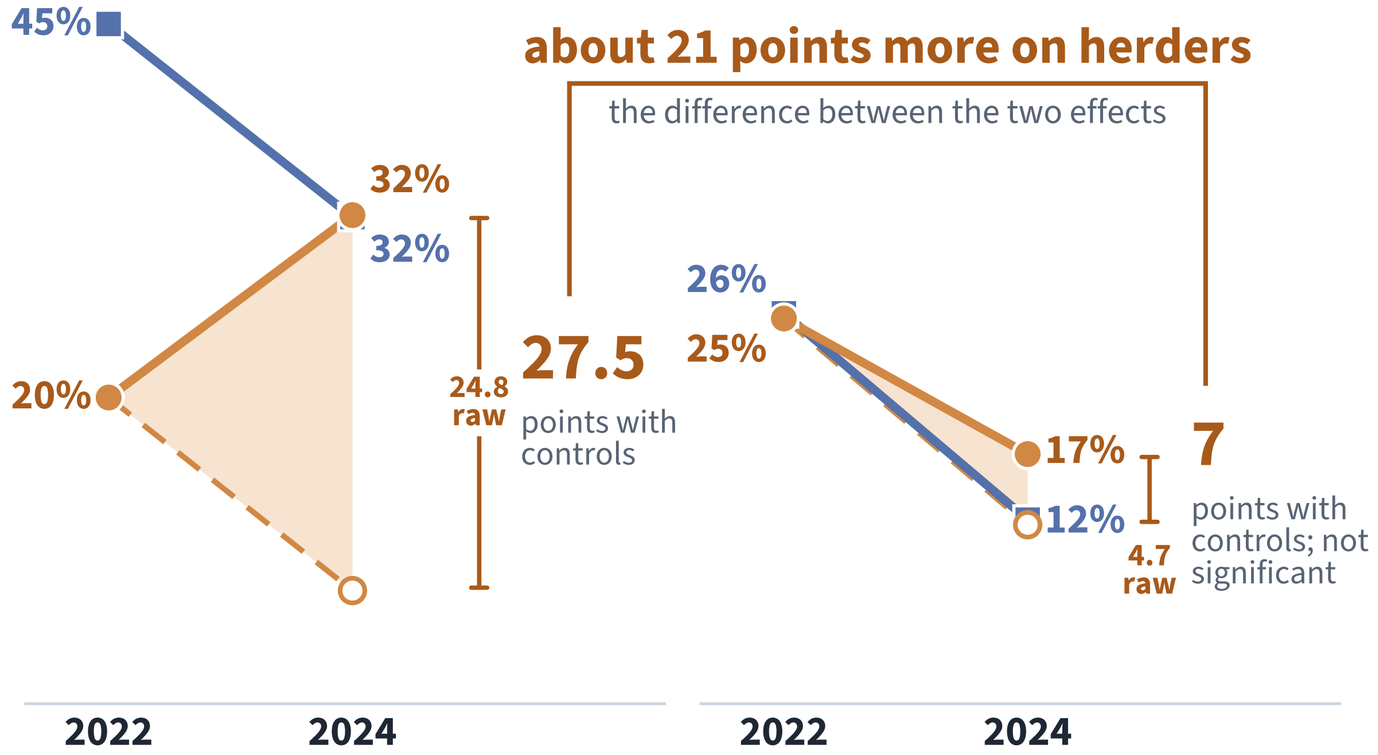

### Who carried it: herders against non-herders

#### Herders

Rising where the winter hit hardest

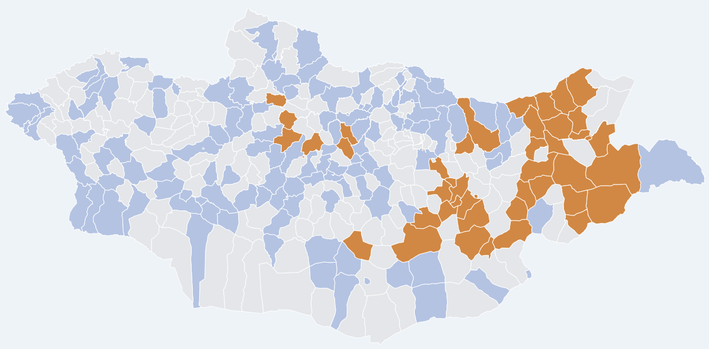
#### Non-herders

Falling less, not significantly




Same households, 2022 and 2024, outside the capital. Dashed: poverty in the severely affected districts had it fallen like the rest. Bracket: the raw gap. Large number: the estimate with controls.

## 1 The design



**Severely affected:** 46 districts, coldest or snowiest against their own history. Lost **27 percent** of livestock.

**Less affected:** 129 others. Lost **9 percent**.

**Difference-in-differences:** the poverty change in **orange** minus **blue** districts, same **2,128 households**.

**Triple difference:** herders against non-herders, there against elsewhere.

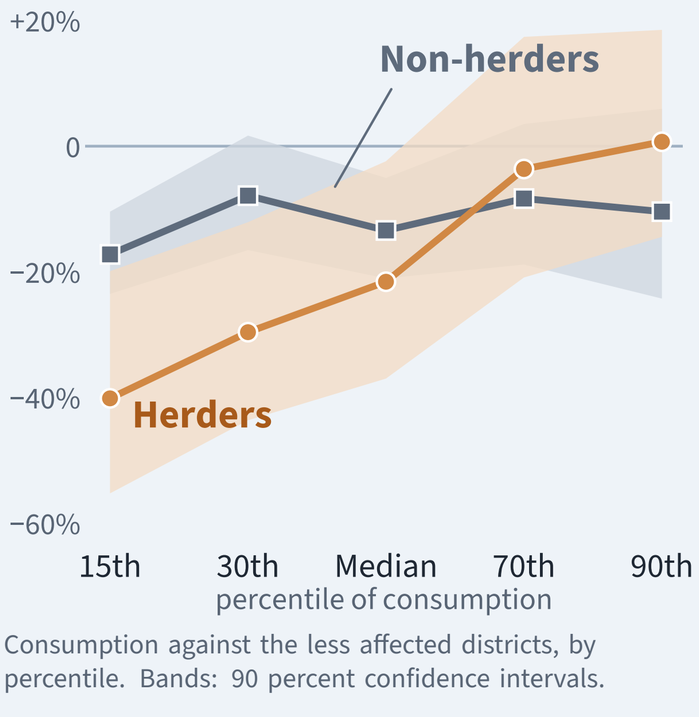



- ✔ Parallel before the winter
- ✔ Placebo winter (2018/19): no effect
- ✔ At least 9.4 points if poorer districts catch up
- ✔ Less affected were also hit: if anything, an understatement

## 2 For whom, and how

### Where the loss landed

Herders crossed the poverty line; poor non-herders fell further below it.




Consumption against the less affected districts, by percentile. Bands: 90 percent confidence intervals.

### Herders: deaths, not distress sales

Herds shrank **36 percent**, against 13 percent elsewhere.

### Non-herders: the food channel

Calories about **12 percent** lower. They had less of their own food as prices rose, and food took 47 percent of poor budgets.

## 3 Policy implications

### What the winter cost

**32,000 to 52,000 people** pushed into or kept in poverty; **$65 to 88 million** of consumption lost in 2024.

### Acting early may cost less

**After:** emergency response
**$15m**

**Before:** $238 per herder household
**$9m**

It must reach the households the winter will hit. In the trial, herds grew for the poorest, not on average.

### Reach beyond herders

Poor non-herders lost food, not herds. Relief aimed at herders misses them.

**A falling national headcount is not evidence that a shock was absorbed.**

Extreme winters will come again.

**When the numbers show progress, look for the hidden cost.**

Poverty rates are the authors' calculations. Data: HSES 2022 and 2024 household panel (National Statistics Office of Mongolia); CRU TS 4.09 temperature; MODIS snow cover; NSO livestock statistics. Grey on the map: no panel households, or the capital. Emergency response: UNDP (2025). Pre-winter transfer: Mogge, Roeckert and Kraehnert (2025, JAERE). Frost texture generated with ChatGPT. The views expressed are the authors' and do not necessarily represent those of the World Bank, its Executive Directors, or the governments they represent.